NATURAL IMAGE STATISTICS FOR SCENE CLASSIFICATION (PHASE-2)

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ABSTRACT

Phase 1 of this project showed that four global statistical features (brightness, contrast, edge density, and spatial correlation) perfectly separate two visually contrasting datasets but collapse to 52.5% accuracy on a six-class scene benchmark, identifying the bottleneck as *representational* rather than algorithmic. This report (Phase 2) tests that diagnosis directly by enriching the handcrafted descriptor with Shannon entropy, four Gray-Level Co-occurrence Matrix (GLCM) texture features, and HSV colour statistics, and by scaling the datasets roughly ten- to twenty-fold. On a balanced 2,000-image binary task (Botswana vs. forest), a Random Forest reaches 89.75% accuracy (87.65% cross-validated), with GLCM homogeneity emerging as one of the most informative feature. On the 6,000-image Intel six-class benchmark, the best model (RBF-SVM) improves to 64.7% from 52.5%, a 12-point gain attributable to the richer representation. Non-parametric significance tests (Mann–Whitney U and Kruskal–Wallis) confirm that most texture and colour descriptors differ significantly across classes. The results validate the Phase 1 hypothesis: enriching the representation, not changing the classifier, drives the improvement, though a clear ceiling remains for visually overlapping categories.

***Index Terms*—** Natural image statistics, GLCM texture features, scene classification, colour features, cross-validation, dimensionality reduction

1. INTRODUCTION

Natural images carry a rich statistical structure, including strong dependencies between neighbouring pixels, characteristic frequency spectra, and structured edge information, that distinguishes them from random noise and underlies much of computer vision [1, 2]. A practical question is how far these low-level regularities, captured as compact handcrafted features, can take a scene-classification system before learned representations become necessary.

The first phase of this project (Phase 1) investigated this question with four global descriptors and two tasks: a binary problem between open savanna (Botswana) and dense forest, and a six-class problem on the Intel scene dataset [3]. The binary task was solved perfectly, but only on a small 100-image dataset, whereas the six-class task plateaued at 52.5%. Crucially, swapping classifiers (Logistic Regression, SVM, Random Forest) barely moved the six-class number, leading to the conclusion that the limiting factor was the *representation* rather than the learning algorithm.

This report (Phase 2) acts on that conclusion. We (i) enrich the feature vector with Shannon entropy, Gray-Level Co-occurrence Matrix (GLCM) texture descriptors [4], and, for the multi-class task, HSV colour statistics; (ii) scale both datasets by an order of magnitude; and (iii) add non-parametric significance testing, PCA and t-SNE visualisation [5], 5-fold cross-validation, feature-importance ranking, and an error study. The aim is to quantify how much of the Phase 1 gap the representation alone can close, while keeping the pipeline fully interpretable.

The remainder of the report is organised as follows. Section 2 states the problem and objectives. Section 3 describes the extended datasets. Section 4 details the methodology. Results are presented in Section 5, discussed in Section 6, and concluded in Section 7.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Phase 1 left an explicit, falsifiable claim: simple handcrafted statistics fail on fine-grained scene classification because they discard texture and colour structure, not because the classifier is too weak. The problem addressed here is to test this claim under conditions where the earlier evaluation was weakest, namely a tiny test set (only 20 binary test images) and a minimal four-dimensional descriptor, and to determine how much performance recovers when the representation is enriched and the data scaled, without resorting to deep networks. The scope remains restricted to interpretable, lightweight features and classical classifiers.

2.2. Objectives

1. To extend the descriptor with entropy, GLCM texture, and HSV colour features, and scale both datasets by roughly 10–20×.
2. To validate each feature with non-parametric tests (Mann–Whitney U, Kruskal–Wallis H).
3. To visualise class separability using PCA and t-SNE.
4. To re-evaluate both tasks with cross-validation, feature-importance ranking, and error analysis.
5. To quantify the gain over Phase 1 and identify the residual bottleneck.

Table 1: Dataset scale: Phase 1 vs. Phase 2.

Task / Dataset	Phase 1	Phase 2
Botswana (binary)	50	1000
Forest (binary)	50	1000
Intel (per class)	100	1000
Intel (total, 6 classes)	600	6000

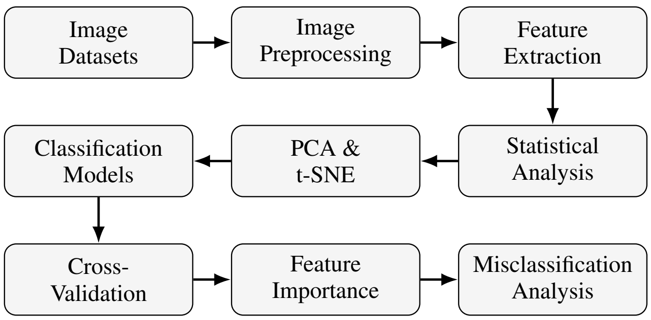


Fig. 1: End-to-end Phase 2 workflow, from raw image datasets through feature extraction, statistical and visual analysis, classification, and post-hoc interpretation.

3. DATASETS

All three datasets from Phase 1 were retained but substantially expanded (Table 1). The two binary classes are visually distinct: Botswana scenes are bright, open and smooth, whereas forest scenes are darker and densely textured.

Binary datasets. The Botswana set, drawn from the UPenn Natural Image Database [6], was extended to 1,000 open-savanna images. The forest set was sourced from a larger public collection (2,626 [7]) images) and randomly subsampled to 1,000 to keep the classes balanced. Compared with Phase 1, the forest class is now far more heterogeneous (varied canopies, lighting, viewpoints), making the task harder and the estimates more reliable.

Intel scene dataset. The Intel Image Classification dataset [3] contains $\sim$25,000 images over six categories (*buildings, forest, glacier, mountain, sea, street*). Phase 2 uses 1,000 images per category (6,000 total), a tenfold increase over Phase 1.

4. METHODOLOGY

The pipeline was implemented in Python (Google Colab) using OpenCV, NumPy, `scikit-image` for texture features [8], and `scikit-learn` for classification and evaluation [9]. The overall analysis pipeline is summarised in Figure 1.

4.1. Feature Extraction

Each image is converted to grayscale and summarised by the descriptor in Table 2. The four Phase 1 statistics are retained, and four GLCM texture measures plus Shannon entropy are added. For the multi-class task, three HSV colour statistics (mean hue, hue standard deviation, mean saturation) are also computed, and images are resized to 150×150 for consistent texture estimation. This yields an **8-dimensional** vector for the binary task and an **11-dimensional** vector for the multi-class task.

Table 2: Handcrafted feature descriptor. Phase 2 additions are marked † (texture/entropy) and ‡ (colour, multi-class only).

Feature	Description
Mean Intensity	Mean grayscale luminance
Std. Deviation	Intensity contrast
Entropy †	Shannon entropy of intensities
Edge Density	Fraction of Canny edge pixels
GLCM Contrast †	Local intensity variation
GLCM Homogeneity †	Closeness of GLCM to diagonal
GLCM Energy †	Textural uniformity
GLCM Correlation †	Linear pixel-pair dependency
Mean Hue ‡	Average HSV hue
Std. Hue ‡	HSV hue variation
Mean Saturation ‡	Average HSV saturation

Whereas the first-order statistics above treat pixels independently, the GLCM captures *second-order* spatial structure: it tabulates how frequently a pair of grey levels co-occurs at a fixed one-pixel offset, so its mass concentrates near the diagonal for smooth regions and disperses for rough texture. Four complementary Haralick descriptors [4] follow: *contrast* measures local roughness, *homogeneity* rewards near-diagonal (uniform) mass, *energy* is large when few grey-level transitions dominate, and *correlation* captures the linear predictability of a pixel from its neighbour. Together they encode the textural regularity that separates dense canopies from smooth savanna. Two scalar descriptors complete the vector: Shannon entropy of the intensity histogram, and edge density (the fraction of Canny edge pixels), both retained from Phase 1.

4.2. Statistical Validation

Because the classes are not Gaussian, feature discriminativeness is assessed with non-parametric tests: the Mann–Whitney U test for the two-class binary problem and the Kruskal–Wallis H test across the six Intel classes, at significance level $\alpha = 0.05$.

4.3. Dimensionality Reduction

Standardised features are projected to two dimensions with PCA (a linear view) and t-SNE [5] (a non-linear manifold view) to visualise class separability.

4.4. Classification and Validation Protocol

Three classifiers are compared: multinomial Logistic Regression, an RBF-kernel SVM, and a Random Forest (200 trees; `max_depth`= 12 for the multi-class case) [10]. Data are split 80/20 with stratification (`random_state`= 42) and standardised to zero mean and unit variance. Each model is additionally evaluated with stratified 5-fold cross-validation, and reported with accuracy, precision, recall, F1-score, and confusion matrices.

5. EXPERIMENTS AND RESULTS

5.1. Experimental Setup

The binary task uses 1,600 training and 400 test images; the multi-class task uses 4,800 training and 1,200 test images, both balanced across classes. All results below refer to the held-out test split unless cross-validation is stated.

Table 3: Binary class-wise feature means (Phase 2).

Feature	Botswana	Forest
Mean Intensity	107.54	85.27
Std. Deviation	47.21	49.26
Entropy	7.13	7.36
Edge Density	0.205	0.245
GLCM Contrast	570.21	672.09
GLCM Homogeneity	0.203	0.121
GLCM Energy	0.0338	0.0167
GLCM Correlation	0.860	0.861

Table 4: Mann–Whitney U test, binary task (sorted by p-value).

Feature	p-value	Significant
Mean Intensity	7.9×10^{-91}	Yes
Entropy	7.3×10^{-44}	Yes
GLCM Energy	4.6×10^{-14}	Yes
GLCM Contrast	1.4×10^{-13}	Yes
Edge Density	2.3×10^{-12}	Yes
Std. Deviation	1.1×10^{-9}	Yes
GLCM Correlation	0.190	No
GLCM Homogeneity	0.452	No

5.2. Binary Feature Analysis

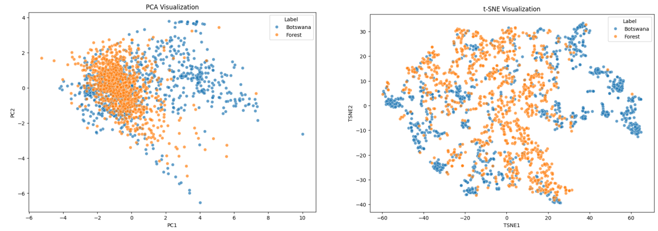
Class-wise feature means (Table 3) show that Botswana scenes are brighter, smoother (higher GLCM homogeneity and energy), and lower in edge density, whereas forest scenes are darker, higher in entropy, and more contrast-rich, as expected for dense vegetation. The Mann–Whitney U test (Table 4) finds six of the eight features significant; only GLCM correlation and GLCM homogeneity are *not* individually significant. The PCA and t-SNE projections (Figure 2) reveal substantial overlap with a clearly separable Botswana tail, visually consistent with a hard but tractable problem, in sharp contrast to the trivial separation seen on the tiny Phase 1 set.

5.3. Binary Classification

Table 5 reports the three classifiers. Random Forest is best at 89.75% test accuracy and 87.65% cross-validated accuracy, followed by SVM (86.5%) and Logistic Regression (79.0%). The low cross-validation standard deviations ($\leq$ 2.3%) indicate stable estimates, unlike the fragile 100% of Phase 1. The feature-importance ranking (Figure 3, left) is led by GLCM homogeneity (0.205) and mean intensity (0.192), confirming that the new texture features carry genuine discriminative weight. Error analysis on the 44 misclassified test images (27 Botswana→forest, 17 forest→Botswana) shows the failures concentrate on darker, textured savanna scenes and sparser forest scenes, i.e. the boundary cases in the overlap region of Figure 2. The confusion matrix is shown in Figure 3 (right).

5.4. Multi-class Feature Analysis

On the Intel dataset, the Kruskal–Wallis H test finds *all* eleven features highly significant ($p < 0.001$). The most discriminative are texture-based: GLCM contrast (H=3178), correlation (H=2986), and homogeneity (H=2770), followed by edge density and mean



(a) PCA (76.5% variance)   (b) t-SNE

Fig. 2: Binary feature-space projections. Forest forms a dense central cluster while Botswana spreads outward; the two classes overlap but are partly separable.

Table 5: Binary classification results (%). CV = 5-fold mean ± std.

Model	Acc.	Prec.	Rec.	F1	CV
Logistic Reg.	79.0	77.8	81.0	79.4	77.6 ± 2.5
SVM (RBF)	86.5	86.5	86.5	86.5	83.2 ± 2.7
Random Forest	**89.75**	91.2	88.0	**89.57**	$\mathbf{87.65 \pm 1.9}$

hue (H=1957), confirming the value of the texture and colour additions. The projections (Figure 4) make the problem structure explicit: forest detaches into its own region, while the other five categories remain heavily entangled.

5.5. Multi-class Classification

Table 6 compares the three classifiers. The RBF-SVM is best at 64.7% test accuracy, ahead of Random Forest (63.5%) and Logistic Regression (61.4%), a 12-point improvement over the Phase 1 figure of 52.5%. The per-class Random Forest report (Table 7) and its confusion matrix (Figure 5) show that forest is recognised almost perfectly (F1 = 0.91), while *buildings* (F1 = 0.49) is the weakest, frequently confused with *street*; *glacier*, *mountain* and *sea* are mutually confused, as predicted by Figure 4.

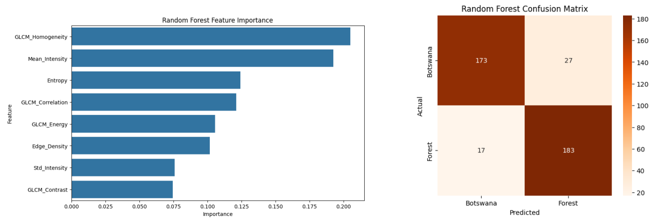
6. DISCUSSION

The Phase 1 diagnosis holds. The headline result is the 52.5% → 64.7% jump on the unchanged six-class benchmark, achieved purely by enriching the representation and scaling the data, with no change to the classifier family. This confirms that the Phase 1 bottleneck was representational; fittingly, GLCM measures top both the Kruskal–Wallis ranking and the binary importance ranking.

Reading the binary "drop" correctly. The binary accuracy falls from the Phase 1 100% to 89.75%, but this is an *improvement* in evidential quality, not a regression. The Phase 1 figure rested on 20 test images from a tiny, clean set; the Phase 2 figure rests on 400 test images from a far more heterogeneous 2,000-image set and is corroborated by cross-validation (87.65 ± 1.9%).

Univariate significance $\neq$ multivariate importance. GLCM homogeneity is *not* individually significant under Mann–Whitney ($p = 0.45$) yet is the *single most important* Random Forest feature. The ensemble exploits interactions that marginal tests cannot see, so feature selection should not rely on univariate tests alone.

Where the ceiling comes from. The residual errors are structural: across all three classifiers the same categories collapse (buildings/street, glacier/sea/mountain), exactly those that overlap in the



(a) RF feature importance

(b) RF confusion matrix

Fig. 3: Random Forest analysis on the binary task: texture features dominate the importance ranking, and errors are balanced across the two classes.

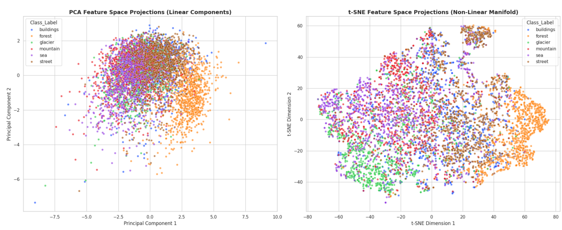


Fig. 4: Multi-class feature-space projections (PCA left, t-SNE right). Forest (orange) separates clearly; the remaining five categories overlap, bounding the achievable accuracy.

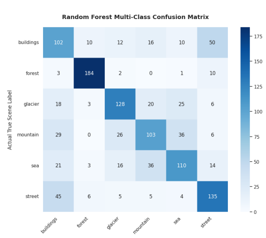


Fig. 5: Random Forest confusion matrix (Intel). Forest is cleanly separated; buildings/street and glacier/sea/mountain dominate the off-diagonal mass.

t-SNE embedding. Global statistics cannot encode the spatial layout that separates, say, a street from a building façade; this natural boundary motivates the future work below.

7. CONCLUSION AND FUTURE WORK

This phase set out to test whether the Phase 1 performance gap was caused by the representation, and to close it without deep learning. Enriching the descriptor with GLCM texture, entropy, and HSV colour features, and scaling the datasets tenfold, raised six-class accuracy from 52.5% to 64.7% and produced a robust, cross-validated 89.75% on a much harder binary task. Texture features were shown, by both significance testing and feature importance, to be the main source of the gain.

The remaining bottleneck is spatial: globally averaged statistics discard the layout information needed to disambiguate similar categories. Promising next steps are local descriptors (e.g. Local Binary Patterns [11], block-wise GLCM), multi-scale frequency analysis, and a Convolutional Neural Network baseline on the same data to measure the handcrafted-vs-learned gap.

Table 6: Multi-class results (%). CV = 5-fold mean accuracy.

Model	CV	Test	Prec.	Rec.	F1
Logistic Reg.	58.8	61.4	60.9	61.4	61.0
Random Forest	60.7	63.5	63.6	63.5	63.4
SVM (RBF)	**62.1**	**64.7**	64.5	64.7	**64.5**

Table 7: Per-class Random Forest performance, Intel dataset.

Class	Precision	Recall	F1
Buildings	0.47	0.51	0.49
Forest	0.89	0.92	**0.91**
Glacier	0.68	0.64	0.66
Mountain	0.57	0.52	0.54
Sea	0.59	0.55	0.57
Street	0.61	0.68	0.64

8. ARTIFACTS AND DEMONSTRATIONS

- **Code Repository** (all Phase 2 notebooks): `https://github.com/akashhh-codes/natural-image-statistics`
- **Intel Dataset:** `https://www.kaggle.com/datasets/puneet6060/intel-image-classification`
- **Botswana Dataset:** `https://web.sas.upenn.edu/upennidb/`
- **Forest Dataset:** `https://images.cv/download/forest/2626?size=size_256&color=color&train=70&val=15`

Note: AI-assisted tools were used for writing and formatting; all analysis, code, and interpretation are the author's own work.

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